Low-Level Design Document

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1. Data Ingestion Module

- Loads raw cryptocurrency data from CSV files
- Ensures correct data types
- Sorts records for time-series consistency

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2. Data Preprocessing Module

- Handles missing values using forward fill
- Removes infinite and invalid values
- Normalizes numeric features

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3. Feature Engineering Module

The following features are generated:

- Log returns for price change measurement
- Rolling volatility for target prediction
- Moving averages for trend analysis
- Liquidity ratios for market activity

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4. Model Training Module

- Splits dataset into training and testing sets
- Applies feature scaling using StandardScaler
- Trains a Random Forest Regressor

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5. Model Evaluation Module

Performance is evaluated using:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- R^2 Score

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6. Conclusion

The low-level design explains internal implementation details and ensures clarity in how each module contributes to volatility prediction.